BCA

Fourth Semester Syllabus 2025-26

University of Mysore

Semester	Course No.	Theory/ Practical	Credits	L-T-P	No. of Hours	Paper Title	Marks	
							SEE	CIE
IV	CAM41T	**Theory**	03	3-0-0	03	Computer Networks	80	20
	CAM41P	**Practical**	02	0-0-2	04	Networking Lab	40	10
	CAM42T	**Theory**	03	3-0-0	03	Python Programming	80	20
	CAM42P	**Practical**	02	0-0-2	04	Python Programming Lab	40	10
	CAM43T	**Theory**	03	3-0-0	03	PHP & MySQL	80	20
	CAM43P	**Practical**	02	0-0-2	04	PHP & MySQL Lab	40	10
	CAE41X	**Theory (Elective1)**	03	3-0-0	03	1. Fundamentals of Data Science 2. Internet of things 3. Software Testing	80	20
	CACP41	**Theory (Compulsory Paper)**	03	2-0-0	02	Digital Marketing	40	10

Semester: IV

Course Code: CAM41T	**Course Title: Computer Networks**
Course Credits: 03 (3-0-0)	**Hours/Week:** 03
Total Contact Hours: 44	**Formative Assessment Marks:** 20
Exam Marks: 80	**Exam Duration:** 03

Course Outcomes (COs):

CO1 Explain digital data transmission between computers.
CO2 Apply data communication and network types in real life.
CO3 Compare layers in networking models.
CO4 Compare protocols in OSI and TCP/IP models.

Course Contents	**Hours**
Unit 1	
Introduction: Characteristics-Delivery, Accuracy, Timeliness and Jitter. **Components-**Message, Sender, Receiver, Transmission medium and protocol, **Topology-**Mesh, Star, Tree, Bus, Ring and Hybrid Topologies. **Transmission modes-** Simplex, Half Duplex, Full Duplex. **Categories of networks** – LAN, WAN, MAN, The ISO/OSI reference model, The TCP/IP reference model. **Digital- to -Analog Conversion:** Amplitude Shift Keying, Frequency Shift Keying, Phase Shift Keying.	11
Unit 2	
Analog- to -Analog Conversion: Amplitude Modulation, Frequency Modulation, Phase Modulation. **The Physical Layer:** Transmission Media – Twisted pair, coaxial cable, optical fiber, radio transmission, microwaves and infrared transmission, Switching – message switching, Multiplexing. Connecting Devices such as Repeater, Hub, Router, Bridge.	11
Unit 3	
The Data Link Layer: Data Link Layer design issues, Error detection – Single parity checking, Checksum, polynomial codes – CRC, Error correction- Hamming code, Elementary data link protocols, sliding window protocols **The Network Layer:** Network layer design issues, Routing algorithms – Dijkstra's shortest path routing. Flooding, Distance vector routing, Hierarchical routing, Link state routing, Congestion, control algorithms – Leaky bucket, token bucket algorithm, admission control, Hop by Hop choke packets.	11
Unit 4	
The Transport Layer and Application Layer: Elements of Transport service, Internet transport protocols (TCP & UDP), Application Layer DNS, Electronic Mailing- Introduction, SMTP, POP3, SNMP, FTP, TELNET and World Wide Web.	11

Text Books:

1. Data Communication & Networking, Behrouza A Forouzan, McGraw Hill.
2. Computer Networks, Andrew S. Tanenbaum, 5th Edition, Pearson Education

Reference Books:

1. Data and Computer Communications, William Stallings, 10th Edition, Pearson Education, 2017.
2. Data Communication and Computer Networks, Brijendra Singh, 3rd Edition, PHI.
3. Data Communication & Network, Dr. Prasad, Wiley Dreamtech.

Course Code: CAM41P	**Course Title: Computer Networks Lab**
Course Credits: 02	**Hours/Week: 04**
Total Contact Hours: 60	**Formative Assessment Marks: 10**
Exam Marks: 40	**Exam Duration: 03**

Course Outcomes (COs):

CO1 Identify and set up basic computer hardware, software, and network configurations.
CO2 Create and test wired network cables and connect networking devices.
CO3 Simulate and configure various network topologies using network simulators.
CO4 Analyze network protocols and services like FTP and wireless LAN through simulation.

Laboratory Program List

Part- A

1. Prepare hardware and software specification for basic computer system and Networking.
2. Study of different types of Network cables and practically implement the cross-wired cable and straight through cable using clamping tool.
3. Identifying the networking devices on a network.
4. Configure the IP address of the computer.
5. Create a basic network and share file and folders.
6. Study of basic network command and Network configuration commands.
7. Installation process of any open-source network simulation software.

Part-B

1. Implement connecting two nodes using network simulator.
2. Implement connecting three nodes considering one node as a central node using network simulator. Implement a network to connect three nodes considering one node as a central node using network simulator
3. Implement bus topology using network simulator.
4. Implement star topology using network simulator.
5. Implement ring topology using network simulator.
6. Demonstrate the use of wireless LAN using network simulator.
7. Implement FTP using TCP bulk transfer using network simulator.
8. Implement connecting multiple routers and nodes and building a Hybrid topology network simulator.

Links for open-source simulation software:
NS3 software: **https://www.nsnam.org/releases/ns-3-30/download/**
Packet Tracer Software: **https://www.netacad.com/courses/packet-tracer**
GNS3 software: **https://www.gns3.com/**

Evaluation Scheme for Lab Examination

Assessment Criteria Marks		**Marks**
Writing	One Program from Part A	15
	One Program from Part B	15
Execution	Any one of the Written Program	5
Viva Voce based on Computer Networks		5
Total		**40**

Semester: IV

Course Code: CAM42T	**Course Title: Python Programming**
Course Credits: 03 (3-0-0)	**Hours/Week:** 03
Total Contact Hours: 44	**Formative Assessment Marks:** 20
Exam Marks: 80	**Exam Duration:** 03

Course Outcomes (COs):

CO1 Demonstrate basic Python programs using control structures and arrays.
CO2 Use functions, strings, lists, and dictionaries in Python.
CO3 Apply OOP, file handling, and exception handling concepts.
CO4 Build GUI apps, work with databases, and visualize data using Python libraries.

Course Contents	**Hours**
Unit 1	
Introduction: Features and Applications of Python; Python Versions; Installation of Python; Python Command Line mode and Python IDEs; Simple Python Program. Identifiers; Keywords; Statements and Expressions; Variables; Operators; Precedence and Association; Data Types; Indentation; Comments; Built-in Functions-Console Input and Console Output, Type Conversions; Python Libraries; Importing Libraries with Examples. **Python Conditional Statement:** Conditional Statements - if, if else, elif, if elif else, match case, looping Statements- while loop, for loop Statement; break, continue statements, range () and exit () functions. **Arrays:** Definition, syntax, accessing the elements of an arrays, array methods.	11
Unit 2	
Python Functions: Types of Functions; Defining Functions, Calling Functions, Passing Parameters/arguments, the return statement; Default Parameters; Command line Arguments; Key Word Arguments; Recursive Functions; Scope and Lifetime of Variables in Functions. **Strings:** Creating and Storing Strings; Accessing Sting Characters; the str() function; Operations on Strings- Concatenation, Comparison, Slicing and Joining, Traversing; Format Specifiers; Escape Sequences; Raw and Unicode Strings; Python String Methods. **Lists:** Creating Lists; Operations on Lists; Built-in Functions on Lists; Implementation of Stacks and Queues using Lists; Nested Lists. **Dictionaries**: Creating Dictionaries; Operations on Dictionaries; Built-in Functions on Dictionaries; Dictionary Methods; Populating and Traversing Dictionaries.	11

Unit 3	
Tuples and Sets: Creating Tuples; Operations on Tuples; Built-in Functions on Tuples; Tuple Methods; Creating Sets; Operations on Sets; Built-in Functions on Sets; Set Methods. **File Handling**: File Types, Operations on Files– Opening and Closing files, Reading and Writing files: write( ) and write lines( ) methods, append() method, read( )and read lines( )methods, with keyword, Splitting words, Renaming and deleting files. **Object Oriented Programming:** Classes and Objects; Creating Classes and Objects; Classes with Multiple Objects; Objects as Arguments; Objects as Return Values; Constructor, types of constructors, Inheritance - Single and Multiple Inheritance, Multilevel and Multipath Inheritance.	11
Unit 4	
GU Interface: The tkinter Module; Window and Widgets; Layout Management-pack, grid and place. **Python SQLite:** The SQLite3 module; SQLite Methods-connect, cursor, execute, close; Connect to Database; Create Table; Operations on Tables- Insert, Select, Update. Delete and Drop Records. **Data Analysis:** NumPy - Introduction to NumPy, Array Creation using NumPy, Operations on Arrays; Pandas- Introduction to Pandas, Series and Data Frames, Creating Data Frames from Excel Sheet and .csv file, Dictionary and Tuples. Operations on Data Frames. **Data Visualization:** Introduction to Data Visualization; Matplotlib Library; Different Types of Charts using Pyplot- Line chart, Bar chart, Histogram & Pie chart.	11

Textbooks

1. **Python Programming:** Using Problem Solving Approach by Reema Thareja, Oxford University Press, 2nd Edition, 2023.
2. **Learning Python** by Mark Lutz, O'Reilly Media, 5th Edition, 2013.

Reference books

1. **Python: The Complete Reference** by Martin C. Brown, McGraw-Hill Education, 2nd Edition, 2018.
2. **Python for Data Analysis by Wes McKinney,** O'Reilly Media, 3rd Edition, 2022.
3. **Head First Python by Paul Barry**, O'Reilly Media, 3rd Edition, 2023.
4. **Advance Core Python Programming,** Meenu Kohli, BPB Publications, 2021.

Course Code: CAM42P	**Course Title: Python Programming Lab**
Course Credits: 02 (0-0-2)	**Hours/Week:** 04
Total Contact Hours: 60	**Formative Assessment Marks:** 10
Exam Marks: 40	**Exam Duration:** 03

Course Outcomes (COs):

CO1 Apply basic Python syntax and logic.
CO2 Use data structures and functions effectively.
CO3 Build apps with Tkinter, SQLite, NumPy, Matplotlib, and Pandas.
CO4 Solve real-world problems using modular Python code.

Laboratory Program List

Part-A

1. Python script for checking the given year is leap year or not.
2. Python script to check if a number belongs to the Fibonacci Sequence
3. Python script to solve Quadratic Equations
4. Python script to display all numbers which are divisible by 7 but are not a multiple of 5, between given range X and Y.
5. Python script to display Multiplication Tables
6. Python script to create a calculator program
7. Explore string functions
8. Implementation of python script that takes a list of words and returns the length of the longest one.
9. Python script handle multiple errors with one except statement.
10. Python script to check whether password is valid or not.
    Conditions for a valid password are:
    - Should have at least one number.
    - Should have at least one uppercase and one lowercase character.
    - Should have at least one special symbol.
    - Should be between 6 to 20 characters long.

Part-B

1. Implement python script to remove duplicates from a list.
2. Implement python script to find the repeated items of a tuple.
3. Implement python script to check whether a given key already exists or not in a dictionary.
4. Write a python script to implement method overloading.
5. Create SQLite Database and Perform Operations on Tables.
6. Create a GUI using Tkinter module.
7. Drawing Line chart and Bar chart using Matplotlib.
8. Drawing Histogram and Pie chart using Matplotlib.
9. Create Array using NumPy and Perform Operations on Array.
10. Create DataFrame from Excel sheet using Pandas and Perform Operations on DataFrames.

Evaluation Scheme for Lab Examination

Assessment Criteria Marks		**Marks**
Writing	One Program from Part A	15
	One Program from Part B	15
Execution	Any one of the Written Program	5
Viva Voce based on Python Programming		5
Total		**40**

Semester: IV

Course Code: CAM43T	**Course Title: PHP & MySQL**
Course Credits: 03 (3-0-0)	**Hours/Week:** 03
Total Contact Hours: 44	**Formative Assessment Marks:** 20
Exam Marks: 80	**Exam Duration:** 03

Course Outcomes (COs):

CO1 Understand PHP basics, syntax, variables, data types, and control structures.
CO2 Use arrays, functions, and strings to create dynamic PHP programs.
CO3 Apply object-oriented concepts and exception handling in PHP.
CO4 Build web applications using forms, sessions, and MySQL with PHP.

Course Contents	Hours
Unit 1	
Introduction to PHP: Introduction to PHP, History and Features of PHP, Installation & Configuration of PHP, Embedding PHP code in Your Web Pages, Understanding PHP, HTML and White Space, Comments in PHP, Sending Data to the Web Browser, Data types, Keywords, Variables, Constants in PHP, Expressions in PHP, Operators in PHP. **Conditional statements:** if, if-else, switch, The? Operator**, Looping statements:** while Loop, do-while Loop, for Loop, foreach loop, break, continue.	11
Unit 2	
Arrays in PHP: Definition, Creating, Accessing Array**, Types of Arrays:** Indexed, Associative arrays, Multidimensional arrays, Accessing Array, Manipulating Arrays, displaying array, Array Functions. **Using Functions in PHP:** Definition, Creating, invoking, user-defined functions, Formal parameters, actual parameters, Function and variable scope, Recursion, Library functions, Date and Time Functions. **Strings in PHP:** Definition, Creating, Declaring, formatting strings, String Functions.	11
Unit 3	
Object Oriented Concepts: Definition, Creation, Declaration, Accessing of Class & Object, Object properties, Object methods, **Constructor:** Definition, Types, Destructor, **Polymorphism:** Method Overloading, Property Overloading. Access Specifiers **Inheritance:** Definition, Single Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Interfaces, Abstract Class, Overriding. **Exception Handling:** try, catch, multi try, multi catch, throw, finally.	11

Unit 4	
Form Handling: Creating HTML Form, Handling HTML Form data in PHP. File Inclusion (Include ( ), Require ( )). **Session Handling:** Definition, session_start (), session_id (), session_destroy () session variables. **Database Handling Using PHP with MySQL:** Introduction to MySQL: Database terms, Data Types. **Accessing MySQL** –Using MySQL Client and Using php MyAdmin, MySQL Commands, Using. PHP MySQL Functions, connecting to MySQL and Selecting the Database, Executing Simple Queries, Retrieving Query Results, Counting Returned Records, Updating Records with PHP	11

Textbooks:

1.**PHP and MySQL Web Development**: Luke Welling, Laura Thomson, Addison-Wesley, 5th Edition, 2016.

2.**Learning PHP, MySQL & JavaScript:** Robin Nixon, O'Reilly Media, 6th Edition, 2021.

Reference Books:

1. **Beginning PHP and MySQL:** From Novice to Professional – W. Jason Gilmore, Apress, 4th Edition, 2010.

Course Code: CAM43P	**Course Title: PHP & MySQL Lab**
Course Credits: 02 (0-0-2)	**Hours/Week:** 04
Total Contact Hours: 60	**Formative Assessment Marks:** 10
Exam Marks: 40	**Exam Duration:** 03

Course Outcomes (COs):

CO1 Implement syntax, control structures, and loop constructs to solve problems.
CO2 Apply arrays, functions, and string operations to build dynamic web features.
CO3 Implement object-oriented programming and exception handling in PHP.
CO4 Develop interactive web applications with form handling, sessions, and MySQL integration.

Laboratory Program List

Part-A

1. Write a PHP script to find the maximum among three given numbers.
2. Write a PHP script to calculate the factorial of a given number.
3. Write a PHP script to check whether a given number is a palindrome.
4. Write a PHP script to reverse a given number and compute the sum of its digits.
5. Write a PHP script to generate a Fibonacci series using a recursive function.
6. Write a PHP script to demonstrate at least seven string functions.
7. Write a PHP script to demonstrate the functionality of date and time functions in PHP.
8. Write a PHP script to insert a new element into an array at a specified position.

Part-B

1. Write a PHP script to demonstrate the use of constructors and destructors in a class.
2. Write a PHP script to demonstrate Multilevel Inheritance.
3. Write a PHP script to demonstrate exception handling by catching a divide-by-zero exception.
4. Write a PHP script to handle form data using the POST method.
5. Write a PHP script to demonstrate session handling by storing and displaying username and user role using session variables.
6. Write a PHP script to create a new database using MySQL and PHP.
7. Write a PHP script to create a table in a MySQL database & insert data using PHP.
8. Develop a PHP application to design a college admission form and store the submitted data into a MySQL database.

Evaluation Scheme for Lab Examination

Assessment Criteria Marks		**Marks**
Writing	One Program from Part A	15
	One Program from Part B	15
Execution	Any one of the Written Program	5
Viva Voce based on PHP & MySQL		5
Total		**40**

Semester: IV

Course Code: CAE4E11 **(Elective)**	**Course Title: Fundamentals of Data Science**
Course Credits: 03(3-0-0)	**Hours/Week:** 03
Total Contact Hours: 44	**Formative Assessment Marks:** 20
Exam Marks: 80	**Exam Duration:** 03

Course Outcomes (COs):

CO1 Explain key concepts of data mining, KDD, and its real-world applications.
CO2 Apply data preprocessing and perform frequent pattern mining using Apriori and FP-Growth.
CO3 Implement and evaluate classification techniques and prediction models.
CO4 Apply clustering methods and evaluate clustering results.

Course Content	Hours
Unit 1	
Data Mining: Introduction, Data Mining Definitions, Steps in Knowledge Discovery in Databases (KDD), Kinds of Data That Can Be Mined in Data Mining , DM functionalities –kinds of patterns can be mined, KDD Vs Data Mining, DBMS Vs Data Mining, DM techniques, Problems, Issues and Challenges in DM, DM applications.	11
Unit 2	
Data Warehouse: Introduction, Definition, Data Warehousing: Three Tier Architecture, Multidimensional Data Model, Schemas for Multidimensional Data Models, OLAP Operations, **Data Cleaning:** Introduction, types of data cleaning tasks(handling missing values, handling noisy data), **Data Integration**: Introduction, Challenges, **Data reduction:** Introduction, Strategies, **Data Transformation:** Introduction, Strategies, Data Transformation by Normalization. **Mining Frequent Patterns:** Basic Concepts(Items, Itemset, Support, Confidence, Maximal Itemset, Closed Itemset), - Association Rule, Mining Association Rules, Frequent Item Set Mining Methods -A priori and Frequent Pattern Growth (FPGrowth) algorithms. Vertical Data Format in Frequent Itemset Mining.	11
Unit 3	
Classification: Basic Concepts, Issues, **Algorithms:** Decision Tree Induction, Attribute Selection Measures, Bayes Classification Methods, Rule-Based Classification, Lazy Learners (or Learning from your Neighbours), k Nearest Neighbour. Metrics for Evaluating Classifier Performance (Precision and Recall), Confusion Matrix.	11
Unit 4	
Clustering: Cluster Analysis, Partitioning Methods (K-Means, K-Medoids), Hierarchical Methods (Algorithmic methods, Probabilistic methods and Bayesian methods), Density-Based Methods (DBSCAN, DENCLUE), Grid-Based Methods (STING, CLIQUE), Dendrogram, Distance Measuring in Algorithmic Methods, Types of Linkage of clusters, Evaluation of Clustering.	11

Text Books:

1. Jiawei Han and Micheline Kamber – “Data Mining Concepts and Techniques” Second Edition Morgan Kaufmann Publishers.

Reference:

1. Arun K Pujari – “Data Mining Techniques” 4th Edition, Universities Press
2. Pang-Ning Tan, Michael Steinbach, Vipin Kumar: Introduction to Data Mining, Pearson Education, 2012.
3. K.P. Soman, Shyam Diwakar, V.Ajay: Insight into Data Mining – Theory and Practice, PHI

Semester: IV

Course Code: CAE4E12 **(Elective)**	**Course Title: Internet of Things**
CourseCredits:03(3-0-0)	**Hours/Week:**03
TotalContactHours:44	**Formative Assessment Marks:** 20
ExamMarks:80	**ExamDuration:**03

Course Outcomes (COs):

CO1 Define key concepts, architecture, and challenges of IoT.
CO2 Explain the role of sensors, actuators, and communication in IoT networks.
CO3 Illustrate the use of IoT protocols and data analytics tools.
CO4 Apply IoT concepts to basic smart city use cases.

Course Content	Hours
Unit 1	
Introduction to IoT: Definition, IoT and Digitization, IoT impact, convergence of IT and OT, IoT Challenges, Comparing IoT Architectures, Core IoT Functional stack, IoT data management and compute stack.	11
Unit 2	
IoT Networks: Sensors Actuators and smart objects, sensor networks, communication criteria, IoT Access Technologies, need for optimization, optimizing IP for IoT.	11
Unit 3	
Application Protocols: The transport Layer, IoT Application Transport Methods, Data Analytics for IoT, Machine Learning, Big data Analytics tools and technology, network analytics.	11
Unit 4	
IoT in Industry: Smart and connected cities – IoT strategy for smarter cities, smart city IoT architecture, smart city security architecture, Smart city use-case examples.	11

Text Book:

1. David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry,"IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Inter of Things" 1st Edition, Pearson Education (Cisco Press Indian Reprint). (ISBN: 9789386873743)

Reference Books:

1. Raj Kamal , " Internet of Things: Architecture and Design", McGraw Hill.2nd edition June 2022.
2. Arsheep Bahga, Vijay Madisetti, Internet Of Things - A Hands-On Approach, Orient Blackswan Private Limited, 2015.

Semester: IV

Course Code: CAE4E13 **(Elective)**	**Course Title: Software Testing**
CourseCredits:03(3-0-0)	**Hours/Week:**03
TotalContactHours:44	**Formative Assessment Marks:** 20
ExamMarks:80	**ExamDuration:**03

Course Outcomes (COs):

CO1 Understand basics of software testing and test case design.
CO2 Apply decision table and data flow testing methods.
CO3 Analyze integration and system testing techniques.
CO4 Evaluate object-oriented and GUI testing approaches.

Course Content	Hours
Unit 1	
Basics of Software Testing and Examples: Basic definitions, Test cases, Insights from a Venn diagram, Identifying test cases, Error and fault taxonomies, Levels of testing, Generalized pseudo code, The triangle problem, The Next Date function, The commission problem, The SATM (Simple Automatic Teller Machine) problem. **Decision Table-Based Testing:** Decision tables, Test cases for the triangle problem, Test cases for the Next Date function, Test cases for the commission problem.	11
Unit 2	
Data Flow Testing: Definition-Use testing, Slice-based testing, Guidelines and observations. **Life Cycle–Based Testing:** Traditional Waterfall Testing**,** Testing in Iterative Life Cycles**,** Agile Testing, **Model-Based Testing:** Testing Based on Models**,** Peterson's Lattice**,** Expressive Capabilities of Mainline Models**,** Modeling Issues**,** Making Appropriate Choices. **Integration Testing:** Introduction, Decomposition-based, call graph- based, Path-based integrations.	11
Unit 3	
System Testing: Definition, Possibilities, Basic concepts for requirements specification, Model-Based Threads, Use Case–Based Threads, ASF (Atomic System Functions). **Object-Oriented Testing:** Units for object-oriented testing, Implications of composition and encapsulation, inheritance, and polymorphism, Levels of object-oriented testing, GUI testing, Dataflow testing for object-oriented software **Class Testing:** Methods as units, Classes as units.	11
Unit 4	
Object-Oriented Integration Testing: UML support for integration testing, MM-paths for object-oriented software, A framework for object-oriented dataflow integration testing. **Object-Oriented System Testing:** Currency converter UML description, UML-based system testing, State chart-based system testing. **GUI Testing:** The currency conversion program, Unit testing, Integration Testing and System testing for the currency conversion program, case study of windshield wiper.	11

Text Books:

1. Paul C. Jorgensen: Software Testing, A Craftsman's Approach, 3rd Edition, Auerbach Publications.

Reference Books:

1. Aditya P Mathur: Foundations of Software Testing, Pearson.
2. Mauro Pezze, Michal Young: Software Testing and Analysis – Process, Principles and Techniques, 1st edition, John Wiley & Sons.
3. Srinivasan Desikan, Gopalaswamy Ramesh: Software testing Principles and Practices.

CIE, SEE and QP Pattern for Theory Courses (3 Credits)

Total Lecture hours per paper: 44
No. of Units 4 (11 Hours Each)
Internal Assessment C1 = 10 Marks, C2 = 10 Marks
Semester End Theory Exam C3 = 80 Marks

Question paper pattern (3 Credits)

Instructions: Answer Part-A and Part-B:

Part-A

Answer any 10 out of 12 Questions (3 Questions drawn from each unit). Each question carries 2 Marks. (10 X 2 =20)
Q. No. 1 to Q. No. 12.

Part-B

Answer all the Questions. Each question carries 15 Marks. (4 X 15 =60)
(Each question with internal choice and with maximum of 3 sub questions)

Semester: IV

Course Code: CACP41	**Course Title: Digital Marketing**
Course Credits: 02(2-0-0)	**Hours/Week:** 02
Total Contact Hours: 30	**Formative Assessment Marks:** 10
Exam Marks: 40	**Exam Duration:** 02

Course Outcomes (COs):

CO1 Describe the basics, evolution, and channels of digital marketing.
CO2 Apply social media and email marketing strategies effectively.
CO3 Create content and mobile marketing plans with analytics.

Course Content	Hours
Unit 1	
Introduction to Digital Marketing: Overview of digital marketing, Evolution of digital marketing, Importance and benefits of digital marketing, Digital marketing channels and platforms **Digital Marketing Strategy and Planning:** Developing a digital marketing strategy, Setting goals and objectives, Budgeting, and resource allocation. Campaign planning and execution, Monitoring and adjusting digital marketing campaigns.	10
Unit 2	
Social Media Marketing: Overview of social media marketing, social media platforms and their features, Creating and optimizing social media profiles, social media content strategy, social media advertising and analytics **Email Marketing:** Introduction to email marketing, building an email list, Creating effective email campaigns, Email automation and segmentation, Email marketing metrics and analytics.	10
Unit 3	
Content Marketing: Understanding content marketing, Content strategy and planning, Content creation and distribution, Content promotion and amplification, Content marketing metrics and analytics. **Mobile Marketing:** Mobile marketing overview, Mobile advertising strategies, Mobile app marketing, Location-based marketing, Mobile marketing analytics.	10

Text Books:

1. "Digital Marketing Strategy: An Integrated Approach to Online Marketing" by Simon Kingsnorth.
2. "Email Marketing Rules: How to Wear a White Hat, Shoot Straight, and Win Hearts" by Chad S. White.

1. "Content Inc.: How Entrepreneurs Use Content to Build Massive Audiences and Create Radically Successful Businesses" by Joe Pulizzi.
2. "Mobile Marketing: How Mobile Technology is Revolutionizing Marketing, Communications and Advertising" by Daniel Rowles.
3. "Web Analytics 2.0: The Art of Online Accountability and Science of Customer Centricity" by Avinash Kaushik.

Scheme of question paper (2 Credits):

Maximum Marks – 40

Part A: Answer any 5 questions: (5x2=10 Marks)

Question No. 1 to 6

Part B: Answer the following questions: (10x3=30 Marks)

7. a) 7. b) or 7. c) 7. d)

8. a) 8. b) or 8. c) 8. d)

9. a) 9. b) or 9. c) 9. d)